

Taylor W. Killian

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RESEARCH INTERESTS

Techniques for efficient and effective *sequential decision making* in the presence of uncertainty: particularly toward improved *reliability* and *generalization*, motivated by real-world challenges

SUBJECT AREAS

- Reinforcement Learning
 - Offline RL
 - Risk-sensitive RL
 - Model-based RL
- LLMs, Reasoning & Agents
- Representation Learning
- Causal Inference
- Approximate Bayesian Inference
- Health and Wellness

EDUCATION

Ph.D., Computer Science **April 2024**
University of Toronto, Toronto, ON, Canada

- GPA: 4.0

Advisor: Marzyeh Ghassemi

Thesis: Clinically Motivated Sequential Decision Making under Uncertainty in Offline Settings

M.Eng, Computational Science and Engineering **May 2017**
Harvard University, Cambridge, MA

- GPA: 3.92

Advisor: Finale Doshi-Velez

Thesis: Robust and Efficient Transfer Learning by Accounting for and Modeling Parameter Variation

BS, Mathematics **April 2013**
Brigham Young University, Provo, Utah

- GPA: 3.83
- President, Mathematics Student Advisory Committee

EXPERIENCE

Assistant Professor, Department of Computer Science **August 2026 - Present**
Brigham Young University

- My research group is focused on developing adaptive decision making algorithms for pragmatic Machine Learning use cases
- Developing Machine Learning curriculum, especially for Reinforcement Learning
- Responsible for BYU CS's Graduate Research Seminar, supporting our Graduate Committee

Principal Scientist (in residence), AI Research **March 2026 - Present**
Lila Sciences

- Providing on-going consulting for post-training efforts of Scientific Reasoning Models
- Developing advantageous research directions in distillation, RL, and temporal abstraction.

Senior Research Scientist, RL and Post-training Lead **February 2025 - February 2026**
MBZUAI, Institute of Foundation Models

- Founding member of North American extension of MBZUAI research entity.
- Developed open-source techniques to facilitate general LLM reasoning capabilities.

- Co-led development of K2-Think, a highly capable 32B parameter reasoning model

Postdoctoral Research Scientist

December 2023 - December 2024

Apple Inc., Special Projects Group – Physical Intelligence

- Established goal-conditioned, contextual RL policies in self-play for multi-agent autonomy.
- Developed solutions for decision-making in partially observed, interactive settings.
- Explored personalization of large-language model agents in long-context settings.

Graduate Research Assistant

August 2019 - April 2024

Department of Computer Science, University of Toronto & Vector Institute

Institute of Medical Engineering and Science, Massachusetts Institute of Technology

- Developed robust representations of patient health, incorporating measures of uncertainty
- Introduced method to identify and avoid detrimental decisions in high-risk environments

Research Intern

May 2023 - September 2023

Apple Inc., AI/ML – Health AI

supervised by Nick Foti

- Developed framework to efficiently adapt global recommendation policies to local contexts
- Instituted agenda to integrate novel and contemporary RL approaches for Apple Health AI

Research Intern

June 2022 - September 2022

Microsoft Research, Health Futures – BioML

supervised by Ava Amini

- Extended distribution-free uncertainty quantification to continuous-time sequential settings
- Developed mechanism to estimate prediction confidence among irregularly sampled time-series

Research Intern

June 2021 - December 2021

Apple Inc., Health AI

supervised by Leon Gatys and Joern-Henrik Jacobsen

- Explored self-supervised learning for representation of user health via physiological signals
- Established state-of-the-art learning architectures for broader use within the Health AI team

Student Researcher / Research Intern

May 2020 - December 2020

Google Research, Brain team

supervised by Marlos Machado and Marc Bellemare

- Studied the utility of embedding uncertainty measures in an RL agent's state representation

Teaching Assistant

August 2019 - May 2020

Department of Computer Science, University of Toronto

- Part of teaching staff for:
 - CSC311 Introduction to Machine Learning
 - CSC2541 Machine Learning for Healthcare
- Developed and administered assignments and exams
- Coordinated projects with clinical collaborators, organized and advised student groups

Associate Technical Staff

June 2017 - July 2019

Air, Missile and Maritime Defense Technology, MIT Lincoln Laboratory

- Led effort to identify and develop areas for Laboratory investment in Artificial Intelligence
- Developed ML algorithms for efficient and accurate performance in low-data regimes

- Fused multiple information sources to reduce false-alarms in aviation passenger screening

Assistant Technical Staff

May 2013 - May 2017

Air, Missile and Maritime Defense Technology, MIT Lincoln Laboratory

- Systems analysis of near- and long-term development of U.S. missile defenses.
- Evaluated the impact of technological and operational improvements to U.S. missile defenses
- Developed and performed data-driven analyses to identify U.S. DoD capability improvements
- Produced briefing materials to present to key U.S. DoD decision makers

Undergraduate Research Assistant

June 2010 - May 2013

Department of Mechanical Engineering, Brigham Young University

- Published research on fluid activated passive dampening as primary author
- Furnished analytical and mathematical support to experimental techniques

Office of Naval Research NREIP Intern

Summer 2011

Naval Surface Warfare Center, Dahlgren, Virginia

- Designed and carried out experiment to study optimal nose shape for submerging projectiles

Language Instructor (Swedish)

December 2009 - June 2010

Missionary Training Center, Church of Jesus Christ of Latter-day Saints

- Prepared lessons to satisfy language and theological curriculum and individual student needs
- Instructed and evaluated language fluency and understanding of concepts.

TEACHING

- CS 270 – Introduction to Machine Learning Brigham Young University, Winter 2027
- CS 401R – Introduction to Reinforcement Learning Brigham Young University, Fall 2026
- CSC 2541 – ML for Healthcare (Lead TA) University of Toronto, Winter 2020
- CSC 311 – Introduction to Machine Learning (TA) University of Toronto, Fall 2019
- CS 282R – RL for Healthcare (Guest Lecture) Harvard University, Fall 2017

PUBLICATIONS

- Gao, C., Li, H., **Killian, T.**, She, J., Wang, R., Ma, L., Cheng, Z., Hao, S., Xu, Z., “**Concise Reasoning in the Lens of Lagrangian Optimization**”. *in Submission*
- Cui, C., **Killian, T.**, Ammanabrolu, P. “**Behavior Cue Reasoning: Monitorable Reasoning Improves Efficiency and Safety through Oversight**”. *in Submission*.
- Deng, M., Hou, J., Sa Neves, L., Pimpalkhute, V., **Killian, T.**, Liu, Z., Xing, E., “**Efficient Agentic Reasoning through Self-Regulated Simulative Planning**”. *in Submission*
- Yu, T., Zhao, K., Chen, C., Hughes, A., **Killian, T.**, Ma, F., Zhang, W., Jenkins, P. “**LARK: Learnability-Grounded Trajectory Selection for Efficient Reasoning Distillation**”. *in Submission*
- Kong, L., Liu, X., Chen, G., Ma, M., Song, X., Sun, Y., Yurochkin, M., **Killian, T.**, Salakhutdinov, R., Zhang, K., Xing, E., Liu, Z. “**From Reasoning Traces to Reusable Modules: Understanding Compositional Generalization in Language Model Reasoning**”. *International Conference on Machine Learning (ICML)*. July 2026.
- Cheng, Z., Xie, Y., Qu, Y., Setlur, A., Hao, S., Pimpalkhute, V., Liang, T., Yao, F., Liu, Z., Xing, E., Smith, V., Salakhutdinov, R., Hu, Z., **Killian, T.**, Kumar, A., “**IsoCompute Playbook: Optimally Scaling Sampling Compute for LLM RL**”. *International Conference on Machine Learning (ICML)*. July 2026.

- Landers, M., Killian, T., Hartvigsen, T, Doryab, A., “**Improving and Accelerating Offline RL in Large Discrete Action Spaces with Structured Policy Initialization**”. *International Conference on Learning Representations (ICLR)*. May 2026.
- Landers, M., Killian, T., Hartvigsen, T, Doryab, A., “**SAINT: Attention-Based Modeling of Sub-Action Dependencies in Multi-Action Policies**”. *arxiv 2505.12109*
- Landers, M., Killian, T., Barnes, H., Hartvigsen, T, Doryab, A., “**BraVE: Offline Reinforcement Learning for Discrete Combinatorial Action Spaces**”. *Advances in Neural Information Processing Systems (NeurIPS)*. December 2025.
- Liu, Z., Tang, L., Jin, L., Li, H., Ranjan, N., Fan, D., Rohatgi, S., Fan, R., Pangarkar, O., Wang, H., Cheng, Z., Sun, S., Han, S., Tan, B., Gosal, G., Han, X., Pimpalkhute, V., Hao, S., Hee, M.S., Hestness, J., Jia, H., Ma, L., Singh, A., Soboleva, D., Vassileva, N., Wang, R., Wu, Y., Sun, Y., Killian, T., Moreno, A., Maggs, J., Ren, H., He, G., Wang, H., Ma, X., Wang, Y., Yurochkin, M., Xing, E. “**K2-v2: A 360-open, reasoning-enhanced LLM**”. *Technical Report*. December 2025. *arxiv 2512.06201*.
- Cheng, Z., Hao, S., Liu, T., Zhou, F., Xie, Y., Yao, F., Bian, Y., Zhuang, Y., Dey, N., Zha, Y., Gu, Y., Zhou, K., Wang, Y., Li, Y., Fan, R., She, J., Gao, C., Saparov, A., Li, H., Killian, T., Yurochkin, M., Liu, Z., Xing, E., Hu, Z., “**Revisiting Reinforcement Learning for LLM Reasoning from A Cross-Domain Perspective**”, *Advances in Neural Information Processing Systems (NeurIPS)*. December 2025.
- Cheng, Z., Fan, R., Hao, S., Killian, T., Li, H., Sun, S., . . . , He, G., Hu, Z., Liu, Z., Xing, E., “**K2-Think: A Parameter-Efficient Reasoning System**”. *Technical Report*. September 2025. *arxiv 2509.07604*
- Cusmano-Towner, M., Hafner, D., Hertzberg, A., Huval, B., Petrenko, A., Vinitzky, E., Wijmans, E., Killian, T., Bowers, S., Sener, O., Krähenbühl, P., Koltun, V., “**Robust Autonomy Emerges from Self-Play**”, *International Conference on Machine Learning (ICML)*. July 2025.
- Killian, T., Zhang, H., Hartvigsen, T., Amini, A., “**Continuous Time Evidential Distributions for Irregular Time Series**”, *arXiv 2307.13503*
- Casper, S., Killian, T., Hadfield-Menell, D., Kreiman, G., “**White-Box Adversarial Policies in Deep Reinforcement Learning**”. *arXiv 2209.02167*
- Killian, T., Parbhoo, S., Ghassemi, M., “**Risk-Sensitive Dead-end Identification in Safety-Critical Offline Reinforcement Learning**”, in *Transactions on Machine Learning Research*. January 2023. <https://openreview.net/forum?id=oK1E0T83gI>.
- Killian, T., Ghassemi, M., Joshi, S., “**Counterfactually Guided Off-policy Transfer in Clinical Settings**”, *Conference on Health, Inference and Learning (CHIL)*. April 2022.
- Fatemi, M., Killian, T., Subramanian, J., Ghassemi, M., “**Medical Dead-ends and Learning to Identify High-Risk States and Treatments**”, *Advances in Neural Information Processing Systems*. December 2021
- Killian, T., Zhang, H., Subramanian, J., Fatemi, M., Ghassemi, M., “**An Empirical Study of Representation Learning for Reinforcement Learning in Healthcare**” *Machine Learning for Health Workshop, NeurIPS*, December 2020
- D’Costa, A., Denkovski, S., Malyska, M., Moon, S.Y., Rufino, B., Yang, Z., Killian, T., Ghassemi, M., “**Multiple Sclerosis Severity Classification From Clinical Text**”, *The 3rd Clinical Natural Language Processing Workshop, EMNLP* 2020.
- Killian, T., Subramanian, J., Fatemi, M., Ghassemi, M., “**Learning Representations for Prediction of Next Patient State**”, *1st Annual ACM Conference on Health, Inference and Learning*, April 2020. **Workshop Spotlight**

- Silva, A., Killian, T., Jimenez, I., Son, S.-H., Gombolay, M. “**Optimization Methods for Interpretable Differentiable Decision Trees Applied to Reinforcement Learning**”, *The 23rd International Conference on Artificial Intelligence and Statistics (AISTATS)*, August 2020.
- Killian, T., Goodwin, J., Brown, O. & Son, S.-H., “**Kernelized Capsule Networks**”, *1st Workshop on Understanding and Improving Generalization in Deep Learning, ICML*, July 2019.
- Yao, J., Killian, T., Konidaris, G. & Doshi-Velez, F., “**Direct Policy Transfer via Hidden Parameter Markov Decision Processes**”, *The 2nd Lifelong Learning: A Reinforcement Learning Approach Workshop, ICML*, July 2018. **Selected for Oral presentation.**
- Jones, A., Killian, T., Hurley, M., & Allen, R., “**Artificial Intelligence and Machine Learning for Decision Support: Recommendations for Investment**”, **Technical Report**, MIT Lincoln Laboratory, June 2018. *Paper not available for public release*
- Killian, T., Daulton, S., Konidaris, G. & Doshi-Velez, F. , “**Robust and Efficient Transfer Learning in Hidden Parameter Markov Decision Processes**”, *Advances in Neural Information Processing Systems*. December 2017 **Selected for an Oral presentation**
- Killian, T., Konidaris, G. & Doshi-Velez, F., “**Robust and Efficient Transfer Learning in Hidden Parameter Markov Decision Processes**”. In *AAAI* (pp. 4949-4950). February 2017.
- Killian, T., Klaus, R. & Truscott, T.T., “**Rebound and jet formation of a fluid-filled sphere**”, *Physics of Fluids* **24**, 122106 (2012), DOI:10.1063/1.4771985.

INVITED TALKS

- *11 August 2026 – Invited Talk, Real World RL Workshop; Hanover, Germany*
“Creative Usage of Rewards for Real World RL”
- *21 January 2026 – Invited Talk, Lila Sciences*
“Fully Open LLM Reasoning at the Frontier”
- *8 May 2025 – Invited Talk, Brigham Young University, Computer Science Department*
“Pragmatic Reinforcement Learning for Real-World Use”
- *19 February 2025 – Guest Lecture, Brigham Young University - Idaho, Data Science Capstone* “Academic and Professional Career Paths for Data Science”
- *4 February 2025 – Invited Talk, Brigham Young University, Math Department*
“Pragmatic Reinforcement Learning in Safety-Critical Domains”
- *3 December 2024 – Invited Talk, Google Research, Health AI*
“Practical Reinforcement Learning in Healthcare”
- *4 November 2024 – Invited Talk, RBC Borealis, HELM Team*
“Pragmatic, Uncertainty Guided Reinforcement Learning”
- *24 October 2024 – Invited Talk, Meta, Applied Reinforcement Learning Team*
“Practical Reinforcement Learning in Safety-Critical Domains”
- *19 April 2024 – Invited Talk, Stanford University – Brunskill Group*
“Pragmatic, Uncertainty Guided Reinforcement Learning”
- *22 September 2023 – Invited Talk, Brown University – bigAI*
“Pragmatic, Uncertainty Guided Reinforcement Learning”
- *13 September 2023 – Invited Talk, NYU Center of Data Science*
“A Pragmatic View of Offline Reinforcement Learning in Safety-Critical Scenarios”
- *22 August 2023 – Invited Talk, Distributional RL Book Launch, MILA*
“Distributional RL for Safety-Critical Offline Settings”

- *20 April 2023 – Invited Talk, Apple, Health AI*
“A Pragmatic View of Reinforcement Learning in Healthcare”
- *2 December 2022 – Invited Talk, NeurIPS 2022 Offline RL Workshop*
“Identification of Dead-ends in Safety-Critical Offline RL”
- *17 November 2022 – Guest Lecture, MIT 6.7950 – RL: Foundations And Methods*
“Context Matters: Leveraging Latent Information to Solve Families of MDPs”
- *10 July 2020 – Invited Talk, Vector Institute Friday Seminar* (virtual)
“Counterfactually Guided Policy Transfer in Clinical Settings”
- *20 February 2020 – Guest Lecture, UofToronto CSC 2541 – ML for Healthcare*
“Reinforcement Learning for Healthcare”
- *18 October 2017 – Guest Lecture, Harvard CS 282R – RL for Healthcare*
“Unwinding the DQN: Tips and Tricks”
- *12 June 2017 – Invited Talk, SMG: Boston, Site Visit*
“Experiences with Computational Science at Harvard”

VOLUNTEER

Program Committee/Reviewer

- **ICML:** 2019-Present
- **NeurIPS:** 2018-Present
- **CHIL:** 2020-Present
- **ICLR:** 2020-Present
- **TMLR:** 2022-Present
- **JMLR:** 2021-Present
- **NeurIPS ML4H Workshop:** 2017-Present
- **NeurIPS Offline RL Workshop:** 2020-Present
- **NeurIPS Deep RL Workshop:** 2020-Present
- **RLC:** 2023-Present

Graduate Committee

Aug 2026 - Present

BYU Computer Science Department

- Organizing weekly Graduate Research Seminar
- Contributing to decisions about adapting Graduate program curriculum and admissions

Program Committee – 2026 RLC Workshop Chair

2025-2026

Organization Committee – 2024 RLC Workshops

2024

“I Can’t Believe It’s Not Better!”: Failure Modes of Sequential Decision-Making in Practice

Finding the Frame: An RLC Workshop for Examining Conceptual Frameworks in RL

- Helped develop the successful proposal of two workshops for the inaugural RL Conference.
- Led ideation and writing of proposals, each focusing on unique points of failure in current research practices in RL
 - First, looking into ideas/concepts that do not scale or otherwise transfer from simulated, toy domains to real-world practice.
 - Second, identifying problematic dogmas in conceptual frameworks for RL and encouraging alternative formulations to unlock new algorithmic capabilities or intended use cases.

Virtualization Chair

2021-2022

Machine Learning for Healthcare Symposium

- Helped lead the development of a standalone symposium, co-located with NeurIPS, in response to growth in interest among the research community
- Led the planning and execution of both a fully virtual event (2021) and hybrid hosting (2022)

Technical Recruiter, Campus Recruiting**December 2014 - July 2019***Human Resources, MIT Lincoln Laboratory*

- Organized campus information events to introduce Laboratory mission and aims
- Served as mentor and advocate for candidates seeking employment

Committee Member; PED Seminar Series**June 2015 - December 2017***MIT Lincoln Laboratory*

- Assisted in the organization and planning of seminar series focused on leveraging modern computation techniques to extract actionable insight
- Hosted leading researchers in Machine Learning and Artificial Intelligence

President, Student Advisory Committee**January 2011-December 2011***Department of Mathematics, Brigham Young University*

- Led effort to improve curriculum and increase collaboration between students and faculty.
- Participated in department review to introduce an applied and computational concentration.
- Planned and carried out activities to promote mathematics and educate BYU community.

Missionary, Sweden Stockholm Mission**March 2007 - March 2009***The Church of Jesus Christ of Latter-day Saints*

- Managed and oversaw the work and safety of 12 other missionaries in remote areas of Sweden.
- Trained 6 newly arrived missionaries in language, culture, and proselyting skills.
- Served full-time as a church representative identifying and meeting the needs of the community.